

# Candidate Pseudolabel Learning: Enhancing Vision-Language Models by Prompt Tuning with Unlabeled Data

Jiahan Zhang, Qi Wei, Feng Liu, Lei Feng

Singapore University of Technology and Design (SUTD)  
Nanyang Technological University (NTU)  
University of Melbourne

May 10, 2026

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## Background: Vision-Language Models are Transforming AI

- VLMs enable zero-shot recognition by aligning images and text.
- Reduce labeling needs across domains and tasks.
- Power applications in search, robotics, healthcare, and remote sensing.

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## Background: Adapting VLMs with Minimal Labels

- Downstream domains differ: fine-grained, remote sensing, textures.
- Prompt tuning adapts VLMs efficiently, but labels cost.
- Unlabeled data is abundant; we need safe adaptation.

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- Model errors propagate as wrong training signals.
- Dominant classes overwhelm rare ones.
- Results: collapse, imbalance, unstable training.

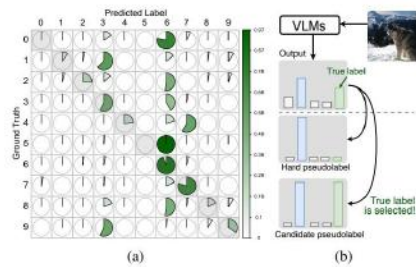


Figure: (a)EuroSAT真标与硬伪标混淆矩阵，显示误差与不平衡；(b)候选伪标为前二高置信类别。

- Textures and fine-grained tasks are brittle.
- Misclassifications are systematic, not random.
- We need balanced, uncertainty-aware signals.

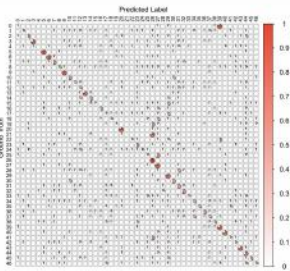


Figure: DTD上真实标签与预测硬伪标签的混淆矩阵

Core Idea: Candidate Pseudolabel Learning (CPL)

- Replace single hard label with a small candidate set.
- Adaptive per-instance size; balanced across classes.
- Optimize with partial-label learning objectives.

Method Overview: From Scores to Candidate Sets

- Build confidence matrix over unlabeled data.
- Dual selection: intra- and inter-instance.
- Intersect to form trainable candidate targets.

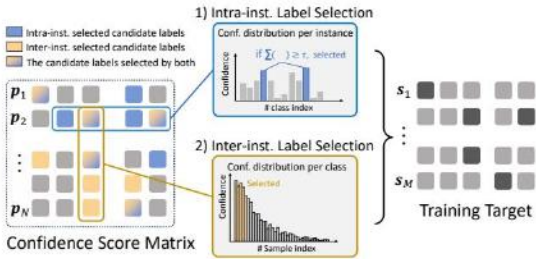


Figure: CPL训练目标生成：由未标注样本置信矩阵 $p$ 构置信矩阵，经层内/层间选取伪标签，形成训练目标 $s$

- Sort class scores per image by confidence.
- Accumulate until threshold  $\tau$  is reached.
- $\tau$  adapts from score distribution via  $\alpha$ .

- Per-class  $\beta$ -quantile filtering by score.
- Keep labels only if within top proportion.
- Mitigates dominance of frequent classes.

- Candidate set = intersection of both selections.
- Convert to multi-hot target vector.
- Train with partial-label losses (CC, RC, CAV, LW).

- Regenerate candidates with updated model.
- Class-wise Top-K per iteration; grow K over time.
- Stabilizes training and improves coverage.

- UL: partial-label loss on unlabeled data.
- SSL: CE on few labels + partial-label loss.
- TRZSL: transductive unlabeled test images + partial-label.

- Nine datasets: Flowers, EuroSAT, CUB, DTD, etc.
- Baselines: FPL, GRIP; plug-in with LaFTer.
- Backbones: ViT-B/32 default; ViT-L/14 explored.

CPL Improves Label Inclusion and Accuracy

- True label inclusion  $\approx 85\%$  vs  $\approx 70\%$  (hard).
- Accuracy rises steadily over iterations.
- Benefits persist throughout training.

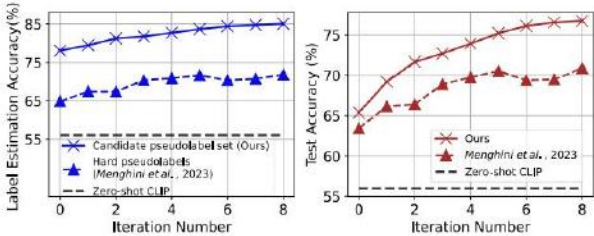


Figure: CPL在RESISC45的标签估计准确率显著优于硬伪标签(Menghini,2023)，并提升测试准确率。

Main Results: Textual Prompt Tuning (Table 1)

- Consistent gains across SSL, UL, TRZSL.
- Large boosts where zero-shot is weak (EuroSAT, DTD).
- Full-unlabeled setting further improves over baselines.

Table: Table 1: Comparison results of top-1 test accuracy (%) on six benchmarks when applying Textual prompts as tuning strategy. Note that “✓” and “×” denote whether full unlabeled data are utilized for fine-tuning or not, respectively.

| Methods        |   | Flowers102 |           |           | RESISC45     |           |           | DTD       |           |           |
|----------------|---|------------|-----------|-----------|--------------|-----------|-----------|-----------|-----------|-----------|
|                |   | SSL        | UL        | TRZSL     | SSL          | UL        | TRZSL     | SSL       | UL        | TRZSL     |
| Zero-shot CLIP |   |            |           |           |              |           |           |           |           |           |
| FPL            | × | 75.960.74  | 65.670.23 | 80.970.00 | 68.130.55    | 63.070.38 | 72.110.00 | 37.105.45 | 44.960.55 | 46.300.03 |
| CPL (Ours)     | × | 77.360.24  | 70.010.21 | 84.600.10 | 71.730.57    | 68.470.34 | 72.160.26 | 54.630.79 | 48.920.17 | 59.791.32 |
| GRIP           | ✓ | 83.600.48  | 69.841.06 | 86.260.00 | 74.110.68    | 70.550.88 | 81.070.00 | 56.070.85 | 46.091.06 | 65.300.01 |
| CPL (Ours)     | ✓ | 89.660.36  | 72.900.78 | 87.350.76 | 80.98.0.11   | 77.390.44 | 85.850.49 | 61.210.56 | 51.910.71 | 68.000.34 |
| CUB            |   | EuroSAT    |           |           | FGVCAircraft |           |           |           |           |           |
| Methods        |   | SSL        | UL        | TRZSL     | SSL          | UL        | TRZSL     | SSL       | UL        | TRZSL     |
| Zero-shot CLIP |   |            |           |           |              |           |           |           |           |           |
| FPL            | × | 55.290.59  | 51.820.00 | 51.570.00 | 32.880.00    | 30.540.00 | 30.540.00 | 17.580.00 | 17.860.00 | 17.860.00 |
| CPL (Ours)     | × | 56.370.45  | 54.180.05 | 64.010.17 | 64.842.15    | 51.451.97 | 54.032.27 | 22.370.66 | 18.900.20 | 28.470.43 |
| GRIP           | ✓ | 56.650.33  | 51.420.21 | 59.480.38 | 58.662.64    | 57.211.77 | 92.330.69 | 16.980.82 | 15.220.71 | 26.080.25 |
| CPL (Ours)     | ✓ | 58.530.24  | 53.470.36 | 66.200.50 | 77.510.80    | 67.260.47 | 93.780.12 | 22.480.63 | 18.350.27 | 30.860.70 |



- CPL module integrates into LaFTer.
- Consistent boosts across datasets in UL.
- Notable gains on EuroSAT (+8.3) and DTD (+10.2).

Table: Table 3: Comparison results of top-1 test accuracy (%) on unsupervised learning when applying parameter-efficient tuning. The best and second-best performances are highlighted via bold and underline, respectively.

|               | Flowers-102 | UCF-101 | CIFAR-100 | EuroSAT | DTD  | CALTECH-101 |
|---------------|-------------|---------|-----------|---------|------|-------------|
| CLIP          | 66.6        | 61.0    | 64.2      | 45.1    | 42.9 | 90.5        |
| CLIP-PR       | 57.7        | 57.9    | 63.2      | 44.2    | 40.1 | 84.8        |
| UPL           | 71.5        | 63.9    | 65.8      | 62.2    | 48.0 | 90.6        |
| LaFTer        | 71.0        | 68.2    | 74.6      | 73.9    | 46.1 | 93.3        |
| LaFTer + Ours | 76.7        | 71.0    | 77.3      | 82.2    | 56.3 | 93.4        |

- TRZSL: strong gains on RESISC45 (e.g., ~86.7%).
- Visual prompts: consistent improvements over GRIP.
- ViT-L/14: CPL scales well with larger encoders.

Why It Works: Better Balance and Inclusion

- Intra-selection reduces dominant-class bias.
- Inter-selection equalizes class exposure.
- Per-class accuracy improves, especially minorities.

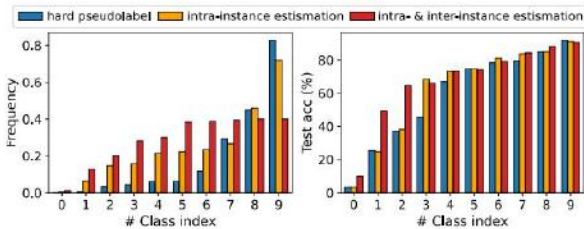


Figure: 左：不同策略伪标签频率分布；右：各策略对EuroSAT类级测试准确率的影响

Candidate Sets Stay Small

- Average size near 1 across datasets.
- Low ambiguity keeps targets sharp.
- Supports stable optimization.

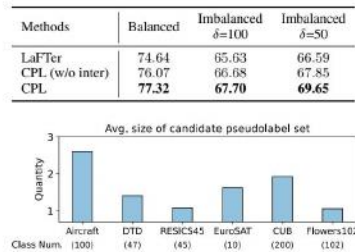


Figure: 文本提示微调UL设置下，6个数据集全体未标注数据伪标签候选平均集合大小可视化

- Good  $\alpha \approx 0.45\text{--}0.75$ ;  $\beta \approx 0.95\text{--}0.97$ .
- Too-high  $\beta$  risks empty sets.
- Performance robust within bands.

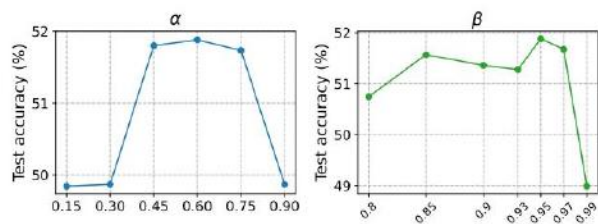


Figure: Hyperparameter study on DTD (UL):  $\beta = 0.95(L)$ ,  $\alpha = 0.60(R)$  fixed

- Grid search shows broad robust plateau.
- Extremes degrade inclusion and coverage.
- Choose moderate  $\alpha$ , high-but-not-extreme  $\beta$ .

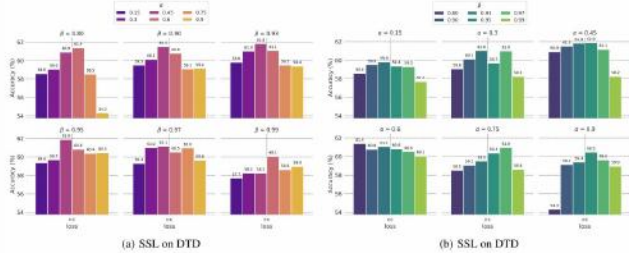


Figure: (a) 不同 $\beta$ 下参数 $\alpha$ 的影响; (b) 不同 $\alpha$ 下参数 $\beta$ 的影响

- Multiple losses work similarly well.
- CC and LW often strongest in UL/SSL.
- Soft CE trails equal-weight candidates.

Table: Table 4: Comparison results of top-1 test accuracy (%) on DTD with textual prompts tuning. The performance of CPL with five different loss functions is evaluated on three tasks.

| Methods        |   | SSL       | UL        | TRZSL     |
|----------------|---|-----------|-----------|-----------|
| Zero-shot CLIP |   | 43.240.00 | 43.450.00 |           |
| FPL            | × | 37.105.45 | 44.960.55 | 46.300.03 |
| CPL Soft CE    | × | 51.830.62 | 47.020.37 | 59.690.59 |
| CPL CC         | × | 54.630.79 | 48.920.17 | 59.791.32 |
| CPL RC         | × | 54.980.49 | 49.960.15 | 59.420.44 |
| CPL CAV        | × | 55.500.29 | 48.690.66 | 59.440.13 |
| CPL LW         | × | 55.210.74 | 49.820.91 | 59.240.72 |
| CPL            | ✓ | 56.070.05 | 46.001.06 | 65.200.01 |

- Unlabeled gains saturate with more data.
- Few-shot labels can beat more unlabeled.
- Especially true for fine-grained tasks.

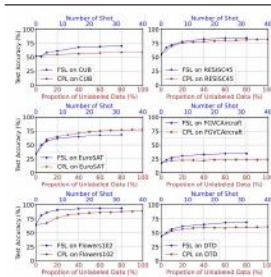


Figure: 文本提示调优下, CLIP+CPL(每类仅2个标注)与全监督小样本提升对比, 从零样本基线起; 蓝/红轴示标注数与未标注占比

- True label may be missing from candidates.
- Extreme  $\beta$  can under-utilize unlabeled data.
- Compute overhead from iterative regeneration.

- Safe, balanced use of unlabeled data for VLMs.
- Transforms pseudolabeling into partial-label learning.
- Consistent gains across datasets, paradigms, backbones.

- Improve candidates when true label excluded.
- Extend to detection, segmentation, retrieval.
- Active data selection and smarter curricula.

- Thank you for your attention!
- Questions and feedback are welcome.

- We thank collaborators and supporting institutions.
- Community resources and open-source tools made this possible.